Membrane Transport and Cellular Excitability

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A.2.1 Transport across cell membranes

Cell membranes regulate exchange between the intracellular and extracellular compartments. Small nonpolar molecules can diffuse through the lipid bilayer, while ions and many polar solutes require channels or carrier proteins. Active transport uses energy directly or indirectly to move substances against their electrochemical gradients.

A.2.2 Osmosis and cell volume

Water moves across permeable membranes in response to differences in effective solute concentration. Changes in extracellular tonicity can therefore alter cell volume, while membrane transporters help maintain the composition of the intracellular environment.

A.2.3 Resting membrane potential

The resting membrane potential reflects ion concentration gradients and the membrane's selective permeability. Potassium conductance is a major determinant in many cells, while transporters maintain the underlying ion gradients.

A.2.4 Action potentials

In excitable cells, depolarization to threshold activates voltage-gated ion channels and produces a rapid, regenerative change in membrane potential. Repolarization and refractory periods shape the timing and propagation of electrical signals.